C Code Lab - Published Questions
Total Questions: 425

1. Phase 1 — Foundation — Introduction to C — Q1: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

2. Phase 1 — Foundation — Structure of a C Program — Q2: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

3. Phase 1 — Foundation — Keywords and Identifiers — Q3: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

4. Phase 1 — Foundation — Variables — Q4: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Swap Two Number Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

5. Phase 1 — Foundation — Data Types — Q5: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium

Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

6. Phase 1 — Foundation — Constants — Q6: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

7. Phase 1 — Foundation — Header Files — Q7: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

8. Phase 1 — Foundation — Input and Output (printf, scanf) — Q8: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

9. Phase 1 — Foundation — Compilation process — Q9: Print Simple Box Border
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Simple Box Border. Read input from stdin and print exact output.
Input Format: One integer n (side length of hollow square drawn with '*'). Middle uses spaces.
Output Format: Print exactly n rows: rows 1 and n are n stars; rows 2..n-1 are star, (n-2 spaces), star.

Constraints: Border size decides rows/columns counts for hollow square artwork.
Sample Input: 3
Expected Output: ***■* *■***
Test Case Count: 3

10. Phase 1 — Foundation — Structure of a C Program — Q10: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

11. Phase 1 — Foundation — Introduction to C — Q11: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

12. Phase 1 — Foundation — Structure of a C Program — Q12: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

13. Phase 1 — Foundation — Keywords and Identifiers — Q13: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Swap Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

14. Phase 1 — Foundation — Variables — Q14: Student Record Variables

Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

15. Phase 1 — Foundation — Data Types — Q15: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

16. Phase 1 — Foundation — Constants — Q16: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

17. Phase 1 — Foundation — Header Files — Q17: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

18. Phase 1 — Foundation — Input and Output (printf, scanf) — Q18: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.

Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

19. Phase 1 — Foundation — Compilation process — Q19: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

20. Phase 1 — Foundation — Keywords and Identifiers — Q20: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

21. Phase 1 — Foundation — Introduction to C — Q21: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

22. Phase 1 — Foundation — Structure of a C Program — Q22: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

23. Phase 1 — Foundation — Keywords and Identifiers — Q23: Check Even/Odd
Module: Phase 1 — Foundation

Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Ch eck Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

24. Phase 1 — Foundation — Variables — Q24: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Name And Ag Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

25. Phase 1 — Foundation — Data Types — Q25: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsign ed char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

26. Phase 1 — Foundation — Constants — Q26: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Add Two Numbers Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

27. Phase 1 — Foundation — Header Files — Q27: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Add Two Numbe s. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.

Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

28. Phase 1 — Foundation — Input and Output (printf, scanf) — Q28: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

29. Phase 1 — Foundation — Compilation process — Q29: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

30. Phase 1 — Foundation — Variables — Q30: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Swap Two Number Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

31. Phase 1 — Foundation — Introduction to C — Q31: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

32. Phase 1 — Foundation — Structure of a C Program — Q32: Add Two Numbers
Module: Phase 1 — Foundation

Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

33. Phase 1 — Foundation — Keywords and Identifiers — Q33: Find Largest Of 3 Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

34. Phase 1 — Foundation — Variables — Q34: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

35. Phase 1 — Foundation — Data Types — Q35: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

36. Phase 1 — Foundation — Constants — Q36: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -

Expected Output: Hello World
Test Case Count: 1

37. Phase 1 — Foundation — Header Files — Q37: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

38. Phase 1 — Foundation — Input and Output (printf, scanf) — Q38: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

39. Phase 1 — Foundation — Compilation process — Q39: Pattern Printing
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Pattern Printing. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print a right-angle star pattern with n lines.
Constraints: One height n determines how many lines of stars.
Sample Input: 3
Expected Output: *■**■***
Test Case Count: 3

40. Phase 1 — Foundation — Data Types — Q40: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

41. Phase 1 — Foundation — Introduction to C — Q41: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Student R

ecord Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

42. Phase 1 — Foundation — Structure of a C Program — Q42: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Sw ap Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

43. Phase 1 — Foundation — Keywords and Identifiers — Q43: Grade Calculator
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Gr ade Calculator. Read input from stdin and print exact output.
Input Format: One integer marks.
Output Format: Print grade: A (>=90), B (>=80), C (>=70), D (>=60), F (<60).
Constraints: One mark from 0 to 100.
Sample Input: 82
Expected Output: B
Test Case Count: 3

44. Phase 1 — Foundation — Variables — Q44: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Character As cii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

45. Phase 1 — Foundation — Data Types — Q45: Check Even/Odd
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Check Even/Odd Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

46. Phase 1 — Foundation — Constants — Q46: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Read Float Decima l Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

47. Phase 1 — Foundation — Header Files — Q47: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Read Float Dec imal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

48. Phase 1 — Foundation — Input and Output (printf, scanf) — Q48: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem s tyle: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

49. Phase 1 — Foundation — Compilation process — Q49: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

50. Phase 1 — Foundation — Constants — Q50: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Print Data Type S izes Lab. Read input from stdin and print exact output.
Input Format: No input.

Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

51. Phase 1 — Foundation — Header Files — Q51: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

52. Phase 1 — Foundation — Input and Output (printf, scanf) — Q52: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

53. Phase 1 — Foundation — Compilation process — Q53: Print Simple Box Border
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Simple Box Border. Read input from stdin and print exact output.
Input Format: One integer n (side length of hollow square drawn with '*'). Middle uses spaces.
Output Format: Print exactly n rows: rows 1 and n are n stars; rows 2..n-1 are star, (n-2 spaces), star.
Constraints: Border size decides rows/columns counts for hollow square artwork.
Sample Input: 3
Expected Output: ***■* *■***
Test Case Count: 3

54. Phase 1 — Foundation — Structure of a C Program — Q54: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai

Expected Output: Riya■Chennai
Test Case Count: 3

55. Phase 1 — Foundation — Introduction to C — Q55: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

56. Phase 1 — Foundation — Structure of a C Program — Q56: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

57. Phase 1 — Foundation — Keywords and Identifiers — Q57: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Swap Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

58. Phase 1 — Foundation — Variables — Q58: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

59. Phase 1 — Foundation — Data Types — Q59: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Read Float Decin

al Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

60. Phase 1 — Foundation — Constants — Q60: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

61. Phase 1 — Foundation — Header Files — Q61: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

62. Phase 1 — Foundation — Input and Output (printf, scanf) — Q62: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

63. Phase 1 — Foundation — Compilation process — Q63: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

64. Phase 1 — Foundation — Keywords and Identifiers — Q64: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

65. Phase 1 — Foundation — Introduction to C — Q65: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

66. Phase 1 — Foundation — Structure of a C Program — Q66: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

67. Phase 1 — Foundation — Keywords and Identifiers — Q67: Check Even/Odd
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

68. Phase 1 — Foundation — Variables — Q68: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Name And Ag Greeting. Read input from stdin and print exact output.

Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

69. Phase 1 — Foundation — Data Types — Q69: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsign ed char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

70. Phase 1 — Foundation — Constants — Q70: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Add Two Numbers Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

71. Phase 1 — Foundation — Header Files — Q71: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Add Two Numbe s. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

72. Phase 1 — Foundation — Input and Output (printf, scanf) — Q72: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem s tyle: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

73. Phase 1 — Foundation — Compilation process — Q73: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

74. Phase 1 — Foundation — Variables — Q74: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Swap Two Number Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

75. Phase 1 — Foundation — Introduction to C — Q75: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

76. Phase 1 — Foundation — Structure of a C Program — Q76: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

77. Phase 1 — Foundation — Keywords and Identifiers — Q77: Find Largest Of 3 Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.

Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

78. Phase 1 — Foundation — Variables — Q78: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

79. Phase 1 — Foundation — Data Types — Q79: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

80. Phase 1 — Foundation — Constants — Q80: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

81. Phase 1 — Foundation — Header Files — Q81: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

82. Phase 1 — Foundation — Input and Output (printf, scanf) — Q82: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

83. Phase 1 — Foundation — Compilation process — Q83: Pattern Printing
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Pattern Printing. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print a right-angle star pattern with n lines.
Constraints: One height n determines how many lines of stars.
Sample Input: 3
Expected Output: *■**■***
Test Case Count: 3

84. Phase 1 — Foundation — Data Types — Q84: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

85. Phase 1 — Foundation — Introduction to C — Q85: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

86. Phase 1 — Foundation — Structure of a C Program — Q86: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Swap Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.

Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

87. Phase 1 — Foundation — Keywords and Identifiers — Q87: Grade Calculator

Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Grade Calculator. Read input from stdin and print exact output.
Input Format: One integer marks.
Output Format: Print grade: A (>=90), B (>=80), C (>=70), D (>=60), F (<60).
Constraints: One mark from 0 to 100.
Sample Input: 82
Expected Output: B
Test Case Count: 3

88. Phase 1 — Foundation — Variables — Q88: Read Character Ascii

Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

89. Phase 1 — Foundation — Data Types — Q89: Check Even/Odd

Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Check Even/Odd Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

90. Phase 1 — Foundation — Constants — Q90: Read Float Decimal Places

Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

91. Phase 1 — Foundation — Header Files — Q91: Read Float Decimal Places

Module: Phase 1 — Foundation

Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

92. Phase 1 — Foundation — Input and Output (printf, scanf) — Q92: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

93. Phase 1 — Foundation — Compilation process — Q93: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

94. Phase 1 — Foundation — Constants — Q94: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

95. Phase 1 — Foundation — Header Files — Q95: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).

Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

96. Phase 1 — Foundation — Input and Output (printf, scanf) — Q96: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

97. Phase 1 — Foundation — Compilation process — Q97: Print Simple Box Border
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Simple Box Border. Read input from stdin and print exact output.
Input Format: One integer n (side length of hollow square drawn with '*'). Middle uses spaces.
Output Format: Print exactly n rows: rows 1 and n are n stars; rows 2..n-1 are star, (n-2 spaces), star.
Constraints: Border size decides rows/columns counts for hollow square artwork.
Sample Input: 3
Expected Output: ***■* *■***
Test Case Count: 3

98. Phase 1 — Foundation — Structure of a C Program — Q98: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

99. Phase 1 — Foundation — Introduction to C — Q99: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

100. Phase 1 — Foundation — Structure of a C Program — Q100: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

101. Phase 1 — Foundation — Keywords and Identifiers — Q101: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Swap Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

102. Phase 1 — Foundation — Variables — Q102: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

103. Phase 1 — Foundation — Data Types — Q103: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

104. Phase 1 — Foundation — Constants — Q104: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.

Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

105. Phase 1 — Foundation — Header Files — Q105: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

106. Phase 1 — Foundation — Input and Output (printf, scanf) — Q106: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

107. Phase 1 — Foundation — Compilation process — Q107: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

108. Phase 1 — Foundation — Keywords and Identifiers — Q108: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

109. Phase 1 — Foundation — Introduction to C — Q109: Print Name And City
Module: Phase 1 — Foundation

Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Print Nam
e And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its
line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

110. Phase 1 — Foundation — Structure of a C Program — Q110: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Re
ad Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

111. Phase 1 — Foundation — Keywords and Identifiers — Q111: Check Even/Odd
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Ch
eck Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

112. Phase 1 — Foundation — Variables — Q112: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Name And Ag
Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

113. Phase 1 — Foundation — Data Types — Q113: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Print Data Type
Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsign
ed char).

Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

114. Phase 1 — Foundation — Constants — Q114: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

115. Phase 1 — Foundation — Header Files — Q115: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

116. Phase 1 — Foundation — Input and Output (printf, scanf) — Q116: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

117. Phase 1 — Foundation — Compilation process — Q117: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

118. Phase 1 — Foundation — Variables — Q118: Swap Two Numbers

Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Swap Two Number Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

119. Phase 1 — Foundation — Introduction to C — Q119: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

120. Phase 1 — Foundation — Structure of a C Program — Q120: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

121. Phase 1 — Foundation — Keywords and Identifiers — Q121: Find Largest Of 3 Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

122. Phase 1 — Foundation — Variables — Q122: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.

Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

123. Phase 1 — Foundation — Data Types — Q123: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Add Two Numbe
Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

124. Phase 1 — Foundation — Constants — Q124: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Print Hello World
. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

125. Phase 1 — Foundation — Header Files — Q125: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Read Character
Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

126. Phase 1 — Foundation — Input and Output (printf, scanf) — Q126: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem s
tyle: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

127. Phase 1 — Foundation — Compilation process — Q127: Pattern Printing
Module: Phase 1 — Foundation
Difficulty: medium

Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Pattern Printing. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print a right-angle star pattern with n lines.
Constraints: One height n determines how many lines of stars.
Sample Input: 3
Expected Output: *■**■***
Test Case Count: 3

128. Phase 1 — Foundation — Data Types — Q128: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Read Character A scii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

129. Phase 1 — Foundation — Introduction to C — Q129: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Student R ecord Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

130. Phase 1 — Foundation — Structure of a C Program — Q130: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Sw ap Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

131. Phase 1 — Foundation — Keywords and Identifiers — Q131: Grade Calculator
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Gr ade Calculator. Read input from stdin and print exact output.
Input Format: One integer marks.
Output Format: Print grade: A (>=90), B (>=80), C (>=70), D (>=60), F (<60).
Constraints: One mark from 0 to 100.
Sample Input: 82
Expected Output: B

Test Case Count: 3

132. Phase 1 — Foundation — Variables — Q132: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

133. Phase 1 — Foundation — Data Types — Q133: Check Even/Odd
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

134. Phase 1 — Foundation — Constants — Q134: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

135. Phase 1 — Foundation — Header Files — Q135: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

136. Phase 1 — Foundation — Input and Output (printf, scanf) — Q136: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.

Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

137. Phase 1 — Foundation — Compilation process — Q137: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

138. Phase 1 — Foundation — Constants — Q138: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

139. Phase 1 — Foundation — Header Files — Q139: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

140. Phase 1 — Foundation — Input and Output (printf, scanf) — Q140: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai

Expected Output: Riya■Chennai
Test Case Count: 3

141. Phase 1 — Foundation — Compilation process — Q141: Print Simple Box Border
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Simple Box Border. Read input from stdin and print exact output.
Input Format: One integer n (side length of hollow square drawn with '*'). Middle uses spaces.
Output Format: Print exactly n rows: rows 1 and n are n stars; rows 2..n-1 are star, (n-2 spaces), star.
Constraints: Border size decides rows/columns counts for hollow square artwork.
Sample Input: 3
Expected Output: ***■* *■***
Test Case Count: 3

142. Phase 1 — Foundation — Introduction to C — Q142: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

143. Phase 1 — Foundation — Structure of a C Program — Q143: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

144. Phase 1 — Foundation — Keywords and Identifiers — Q144: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Swap Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

145. Phase 1 — Foundation — Variables — Q145: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Student Record Va

riables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

146. Phase 1 — Foundation — Data Types — Q146: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Read Float Decim
al Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

147. Phase 1 — Foundation — Constants — Q147: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Read Character As
cii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

148. Phase 1 — Foundation — Header Files — Q148: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Print Hello Wo
rld. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

149. Phase 1 — Foundation — Input and Output (printf, scanf) — Q149: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem s
tyle: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

150. Phase 1 — Foundation — Compilation process — Q150: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

151. Phase 1 — Foundation — Introduction to C — Q151: Print Name And City
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Print Name And City. Read input from stdin and print exact output.
Input Format: Line 1: name. Line 2: city.
Output Format: Print name on line 1, city on line 2.
Constraints: You get two answers the user typed: first the name word, second the city word — each alone on its line.
Sample Input: Riya■Chennai
Expected Output: Riya■Chennai
Test Case Count: 3

152. Phase 1 — Foundation — Structure of a C Program — Q152: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

153. Phase 1 — Foundation — Keywords and Identifiers — Q153: Check Even/Odd
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

154. Phase 1 — Foundation — Variables — Q154: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Name And Ag
Greeting. Read input from stdin and print exact output.

Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

155. Phase 1 — Foundation — Data Types — Q155: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsign ed char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

156. Phase 1 — Foundation — Constants — Q156: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Add Two Numbers Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

157. Phase 1 — Foundation — Header Files — Q157: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Add Two Numbe s. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

158. Phase 1 — Foundation — Input and Output (printf, scanf) — Q158: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem s tyle: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

159. Phase 1 — Foundation — Compilation process — Q159: Print Data Type Sizes Lab
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Print Data Type Sizes Lab. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print four integers separated by spaces: sizeof(int) sizeof(float) sizeof(double) sizeof(unsigned char).
Constraints: No input values (lab assumes fixed widths for grading).
Sample Input: -
Expected Output: 4 4 8 1
Test Case Count: 1

160. Phase 1 — Foundation — Introduction to C — Q160: Read Name And Age Greeting
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Read Name And Age Greeting. Read input from stdin and print exact output.
Input Format: Line 1: name string. Line 2: integer age.
Output Format: Print one greeting sentence exactly.
Constraints: Person gives one word nickname, then a normal human age.
Sample Input: Alex■21
Expected Output: Hello Alex, you are 21 years old.
Test Case Count: 3

161. Phase 1 — Foundation — Structure of a C Program — Q161: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

162. Phase 1 — Foundation — Keywords and Identifiers — Q162: Find Largest Of 3 Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

163. Phase 1 — Foundation — Variables — Q163: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.

Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

164. Phase 1 — Foundation — Data Types — Q164: Add Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

165. Phase 1 — Foundation — Constants — Q165: Print Hello World
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Print Hello World. Read input from stdin and print exact output.
Input Format: No input.
Output Format: Print exactly: Hello World
Constraints: This program runs without waiting for typed input.
Sample Input: -
Expected Output: Hello World
Test Case Count: 1

166. Phase 1 — Foundation — Header Files — Q166: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

167. Phase 1 — Foundation — Input and Output (printf, scanf) — Q167: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

168. Phase 1 — Foundation — Compilation process — Q168: Pattern Printing
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Pattern Printing. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print a right-angle star pattern with n lines.
Constraints: One height n determines how many lines of stars.
Sample Input: 3
Expected Output: *■**■***
Test Case Count: 3

169. Phase 1 — Foundation — Introduction to C — Q169: Student Record Variables
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Introduction to C'.■Problem style: Student Record Variables. Read input from stdin and print exact output.
Input Format: Four lines: name, age, grade letter, tuition fees.
Output Format: Print four fields separated by spaces: name age grade fees.
Constraints: Four separate answers: word name, numeric age, one letter grade, tuition digits.
Sample Input: Mia■19■B■72000
Expected Output: Mia 19 B 72000
Test Case Count: 3

170. Phase 1 — Foundation — Structure of a C Program — Q170: Swap Two Numbers
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Structure of a C Program'.■Problem style: Swap Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped values: b a.
Constraints: Two integers arrive in order.
Sample Input: 7 3
Expected Output: 3 7
Test Case Count: 3

171. Phase 1 — Foundation — Keywords and Identifiers — Q171: Grade Calculator
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Keywords and Identifiers'.■Problem style: Grade Calculator. Read input from stdin and print exact output.
Input Format: One integer marks.
Output Format: Print grade: A (>=90), B (>=80), C (>=70), D (>=60), F (<60).
Constraints: One mark from 0 to 100.
Sample Input: 82
Expected Output: B
Test Case Count: 3

172. Phase 1 — Foundation — Variables — Q172: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Variables'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.

Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

173. Phase 1 — Foundation — Data Types — Q173: Check Even/Odd
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Data Types'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

174. Phase 1 — Foundation — Constants — Q174: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Constants'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

175. Phase 1 — Foundation — Header Files — Q175: Read Float Decimal Places
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Header Files'.■Problem style: Read Float Decimal Places. Read input from stdin and print exact output.
Input Format: One float on one line.
Output Format: Print three values with 2, 4, then 6 decimal places, space-separated.
Constraints: One fractional number shown with decimal digits.
Sample Input: 3.14159
Expected Output: 3.14 3.1416 3.141590
Test Case Count: 3

176. Phase 1 — Foundation — Input and Output (printf, scanf) — Q176: Read Character Ascii
Module: Phase 1 — Foundation
Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Input and Output (printf, scanf)'.■Problem style: Read Character Ascii. Read input from stdin and print exact output.
Input Format: A single character on one line.
Output Format: Print its decimal ASCII code.
Constraints: One visible character keyed by user.
Sample Input: A
Expected Output: 65
Test Case Count: 3

177. Phase 1 — Foundation — Compilation process — Q177: Add Two Numbers
Module: Phase 1 — Foundation

Difficulty: medium
Description: Solve a phase 1 — foundation coding task focused on 'Compilation process'.■Problem style: Add Two Numbers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print a + b.
Constraints: The user enters two ordinary integers.
Sample Input: 7 3
Expected Output: 10
Test Case Count: 3

178. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q1: Check d
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, Logical, Assignment'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

179. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q2: Print Multiplic ble
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

180. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q3: Grade Cal r
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (if, else, switch)'.■Problem style: Grade Calculator. Read input from stdin and print exact output.
Input Format: One integer marks.
Output Format: Print grade: A (>=90), B (>=80), C (>=70), D (>=60), F (<60).
Constraints: One mark from 0 to 100.
Sample Input: 82
Expected Output: B
Test Case Count: 3

181. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q4: Print Multiplication Ta
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.

Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

182. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q5: Break On Sum

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, continue, goto)'.■Problem style: Break On Negative Sum. Read input from stdin and print exact output.
Input Format: First line count n, then n integers one per line (last may be negative).
Output Format: Print sum of values before first negative integer.
Constraints: Count prefixed then sequence entries until negative sentinel occurs.
Sample Input: 3■5■8■-2
Expected Output: 13
Test Case Count: 3

183. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q6: Print Multiplic ble

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

184. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q7: Find L Of 3 Numbers

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, Logical, Assignment'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

185. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q8: Read Until Ze

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52

Test Case Count: 3

186. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q9: Day Of We tch

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Day Of Week Switch. Read input from stdin and print exact output.
Input Format: One integer.
Output Format: Print abbreviated weekday exactly: Mon Tue Wed Thu Fri Sat Sun
Constraints: Integer 1..7 referencing weekday numbering from handout.
Sample Input: 3
Expected Output: Wed
Test Case Count: 3

187. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q10: Read Until Zero Sum

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

188. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q11: Read Ur Sum

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

189. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q12: Grade Ca or

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Grade Calculator. Read input from stdin and print exact output.
Input Format: One integer marks.
Output Format: Print grade: A (>=90), B (>=80), C (>=70), D (>=60), F (<60).
Constraints: One mark from 0 to 100.
Sample Input: 82
Expected Output: B
Test Case Count: 3

190. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q13: Simp est

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, Logical, Assignment'.■Problem style: Simple Interest. Read input from stdin and print exact output.
Input Format: Three lines: principal P, rate R, time T (all integers).
Output Format: Print simple interest as integer.
Constraints: Three integers Principal Rate Time whole numbers.
Sample Input: 1000■5■2
Expected Output: 100
Test Case Count: 3

191. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q14: Find Largest ers
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

192. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q15: Check Ev
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (if, else, switch)'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

193. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q16: Prime Check
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Prime Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Prime or Not
Constraints: One integer roughly ≥2 check primality semantics.
Sample Input: 7
Expected Output: Prime
Test Case Count: 3

194. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q17: Calculat h Menu
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, continue, goto)'.■Problem style: Calculator Switch Menu. Read input from stdin and print exact output.
Input Format: Line 1: op. Line 2: a. Line 3: b. Integer division for '/'.

Output Format: Print integer result.
Constraints: Operator letter then two operands lines.
Sample Input: +■4■11
Expected Output: 15
Test Case Count: 3

195. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q18: Print Multiplication Ta
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

196. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q19: Calcu witch Menu
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Calculator Switch Menu. Read input from stdin and print exact output.
Input Format: Line 1: op. Line 2: a. Line 3: b. Integer division for '/'.
Output Format: Print integer result.
Constraints: Operator letter then two operands lines.
Sample Input: +■4■11
Expected Output: 15
Test Case Count: 3

197. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q20: Break On Ne
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Break On Negative Sum. Read input from stdin and print exact output.
Input Format: First line count n, then n integers one per line (last may be negative).
Output Format: Print sum of values before first negative integer.
Constraints: Count prefixed then sequence entries until negative sentinel occurs.
Sample Input: 3■5■8■-2
Expected Output: 13
Test Case Count: 3

198. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q21: Prime Ch
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Prime Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Prime or Not
Constraints: One integer roughly ≥2 check primality semantics.
Sample Input: 7
Expected Output: Prime
Test Case Count: 3

199. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q22: Fibonacci Series
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Fibonacci Series. Read input from stdin and print exact output.
Input Format: One integer n (number of terms).
Output Format: Print first n Fibonacci numbers separated by spaces.
Constraints: Count of terms to emit.
Sample Input: 7
Expected Output: 0 1 1 2 3 5 8
Test Case Count: 3

200. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q23: Print Mu tion Table
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, continue, goto)'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

201. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q24: Break O e Sum
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, continue, goto)'.■Problem style: Break On Negative Sum. Read input from stdin and print exact output.
Input Format: First line count n, then n integers one per line (last may be negative).
Output Format: Print sum of values before first negative integer.
Constraints: Count prefixed then sequence entries until negative sentinel occurs.
Sample Input: 3■5■8■-2
Expected Output: 13
Test Case Count: 3

202. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q25: Print lication Table
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, Logical, Assignment'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

203. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q26: Check Even/
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o

perators'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

204. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q27: Leap Yea
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Leap Year. Read input from stdin and print exact output.
Input Format: One integer year.
Output Format: Print Leap or Not
Constraints: One calendar year typed.
Sample Input: 2024
Expected Output: Leap
Test Case Count: 3

205. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q28: Factorial Of Number
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Factorial Of Number. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print n!.
Constraints: Small non-negative integer n (factorial grows fast).
Sample Input: 5
Expected Output: 120
Test Case Count: 3

206. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q29: Simple I
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Simple Interest. Read input from stdin and print exact output.
Input Format: Three lines: principal P, rate R, time T (all integers).
Output Format: Print simple interest as integer.
Constraints: Three integers Principal Rate Time whole numbers.
Sample Input: 1000■5■2
Expected Output: 100
Test Case Count: 3

207. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q30: Print Multipl able
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50

Test Case Count: 3

208. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q31: Find Of 3 Numbers

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, Logical, Assignment'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

209. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q32: Read Until Z

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

210. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q33: Day Of W itch

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (if, else, switch)'.■Problem style: Day Of Week Switch. Read input from stdin and print exact output.
Input Format: One integer.
Output Format: Print abbreviated weekday exactly: Mon Tue Wed Thu Fri Sat Sun
Constraints: Integer 1..7 referencing weekday numbering from handout.
Sample Input: 3
Expected Output: Wed
Test Case Count: 3

211. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q34: Read Until Zero Sum

Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

212. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q35: Read Ur Sum

Module: Phase 2 — Logic Building & Flow Control

Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

213. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q36: Grade Ca or
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Grade Calculator. Read input from stdin and print exact output.
Input Format: One integer marks.
Output Format: Print grade: A (>=90), B (>=80), C (>=70), D (>=60), F (<60).
Constraints: One mark from 0 to 100.
Sample Input: 82
Expected Output: B
Test Case Count: 3

214. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q37: Simp est
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Simple Interest. Read input from stdin and print exact output.
Input Format: Three lines: principal P, rate R, time T (all integers).
Output Format: Print simple interest as integer.
Constraints: Three integers Principal Rate Time whole numbers.
Sample Input: 1000■5■2
Expected Output: 100
Test Case Count: 3

215. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q38: Find Largest ers
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

216. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q39: Check E
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.

Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

217. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q40: Prime Check
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Prime Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Prime or Not
Constraints: One integer roughly ≥2 check primality semantics.
Sample Input: 7
Expected Output: Prime
Test Case Count: 3

218. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q41: Calcula h Menu
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, continue, goto)'.■Problem style: Calculator Switch Menu. Read input from stdin and print exact output.
Input Format: Line 1: op. Line 2: a. Line 3: b. Integer division for '/'.
Output Format: Print integer result.
Constraints: Operator letter then two operands lines.
Sample Input: +■4■11
Expected Output: 15
Test Case Count: 3

219. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q42: Print Multiplication T
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

220. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q43: Calc witch Menu
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, Logical, Assignment'.■Problem style: Calculator Switch Menu. Read input from stdin and print exact output.
Input Format: Line 1: op. Line 2: a. Line 3: b. Integer division for '/'.
Output Format: Print integer result.
Constraints: Operator letter then two operands lines.
Sample Input: +■4■11
Expected Output: 15
Test Case Count: 3

221. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q44: Break On Ne
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Break On Negative Sum. Read input from stdin and print exact output.
Input Format: First line count n, then n integers one per line (last may be negative).
Output Format: Print sum of values before first negative integer.
Constraints: Count prefixed then sequence entries until negative sentinel occurs.
Sample Input: 3■5■8■-2
Expected Output: 13
Test Case Count: 3

222. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q45: Prime Ch
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (if, else, switch)'.■Problem style: Prime Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Prime or Not
Constraints: One integer roughly ≥2 check primality semantics.
Sample Input: 7
Expected Output: Prime
Test Case Count: 3

223. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q46: Fibonacci Series
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Fibonacci Series. Read input from stdin and print exact output.
Input Format: One integer n (number of terms).
Output Format: Print first n Fibonacci numbers separated by spaces.
Constraints: Count of terms to emit.
Sample Input: 7
Expected Output: 0 1 1 2 3 5 8
Test Case Count: 3

224. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q47: Print Mu tion Table
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, continue, goto)'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

225. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q48: Break O e Sum
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c

ontinue, goto)'.■Problem style: Break On Negative Sum. Read input from stdin and print exact output.
Input Format: First line count n, then n integers one per line (last may be negative).
Output Format: Print sum of values before first negative integer.
Constraints: Count prefixed then sequence entries until negative sentinel occurs.
Sample Input: 3■5■8■-2
Expected Output: 13
Test Case Count: 3

226. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q49: Print lication Table
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

227. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q50: Check Even/
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

228. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q51: Leap Yea
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Leap Year. Read input from stdin and print exact output.
Input Format: One integer year.
Output Format: Print Leap or Not
Constraints: One calendar year typed.
Sample Input: 2024
Expected Output: Leap
Test Case Count: 3

229. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q52: Factorial Of Number
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Factorial Of Number. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print n!.
Constraints: Small non-negative integer n (factorial grows fast).
Sample Input: 5
Expected Output: 120

Test Case Count: 3

230. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q53: Simple I
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Simple Interest. Read input from stdin and print exact output.
Input Format: Three lines: principal P, rate R, time T (all integers).
Output Format: Print simple interest as integer.
Constraints: Three integers Principal Rate Time whole numbers.
Sample Input: 1000■5■2
Expected Output: 100
Test Case Count: 3

231. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q54: Print Multipl able
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

232. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q55: Find Of 3 Numbers
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

233. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q56: Read Until Z
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

234. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q57: Day Of W itch
Module: Phase 2 — Logic Building & Flow Control

Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Day Of Week Switch. Read input from stdin and print exact output.
Input Format: One integer.
Output Format: Print abbreviated weekday exactly: Mon Tue Wed Thu Fri Sat Sun
Constraints: Integer 1..7 referencing weekday numbering from handout.
Sample Input: 3
Expected Output: Wed
Test Case Count: 3

235. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q58: Read Until Zero Sum
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

236. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q59: Read Un Sum
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

237. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q60: Grade Ca or
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Grade Calculator. Read input from stdin and print exact output.
Input Format: One integer marks.
Output Format: Print grade: A (>=90), B (>=80), C (>=70), D (>=60), F (<60).
Constraints: One mark from 0 to 100.
Sample Input: 82
Expected Output: B
Test Case Count: 3

238. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q61: Simp est
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Simple Interest. Read input from stdin and print exact output.
Input Format: Three lines: principal P, rate R, time T (all integers).

Output Format: Print simple interest as integer.
Constraints: Three integers Principal Rate Time whole numbers.
Sample Input: 1000■5■2
Expected Output: 100
Test Case Count: 3

239. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q62: Find Largest ers
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

240. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q63: Check Ev
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (if, else, switch)'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

241. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q64: Prime Check
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Prime Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Prime or Not
Constraints: One integer roughly ≥2 check primality semantics.
Sample Input: 7
Expected Output: Prime
Test Case Count: 3

242. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q65: Calculat h Menu
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, continue, goto)'.■Problem style: Calculator Switch Menu. Read input from stdin and print exact output.
Input Format: Line 1: op. Line 2: a. Line 3: b. Integer division for '/'.
Output Format: Print integer result.
Constraints: Operator letter then two operands lines.
Sample Input: +■4■11
Expected Output: 15
Test Case Count: 3

243. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q66: Print Multiplication Ta
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

244. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q67: Calcu witch Menu
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Calculator Switch Menu. Read input from stdin and print exact output.
Input Format: Line 1: op. Line 2: a. Line 3: b. Integer division for '/'.
Output Format: Print integer result.
Constraints: Operator letter then two operands lines.
Sample Input: +■4■11
Expected Output: 15
Test Case Count: 3

245. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q68: Break On Ne
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Break On Negative Sum. Read input from stdin and print exact output.
Input Format: First line count n, then n integers one per line (last may be negative).
Output Format: Print sum of values before first negative integer.
Constraints: Count prefixed then sequence entries until negative sentinel occurs.
Sample Input: 3■5■8■-2
Expected Output: 13
Test Case Count: 3

246. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q69: Prime Ch
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Prime Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Prime or Not
Constraints: One integer roughly ≥2 check primality semantics.
Sample Input: 7
Expected Output: Prime
Test Case Count: 3

247. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q70: Fibonacci Series
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Fibonacci Series. Read input from stdin and print exact output.

Input Format: One integer n (number of terms).
Output Format: Print first n Fibonacci numbers separated by spaces.
Constraints: Count of terms to emit.
Sample Input: 7
Expected Output: 0 1 1 2 3 5 8
Test Case Count: 3

248. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q71: Print Mu tion Table
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

249. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q72: Break O e Sum
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Break On Negative Sum. Read input from stdin and print exact output.
Input Format: First line count n, then n integers one per line (last may be negative).
Output Format: Print sum of values before first negative integer.
Constraints: Count prefixed then sequence entries until negative sentinel occurs.
Sample Input: 3■5■8■-2
Expected Output: 13
Test Case Count: 3

250. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q73: Print lication Table
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

251. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q74: Check Even/
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8

Expected Output: Even
Test Case Count: 3

252. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q75: Leap Yea
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Leap Year. Read input from stdin and print exact output.
Input Format: One integer year.
Output Format: Print Leap or Not
Constraints: One calendar year typed.
Sample Input: 2024
Expected Output: Leap
Test Case Count: 3

253. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q76: Factorial Of Number
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Factorial Of Number. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print n!.
Constraints: Small non-negative integer n (factorial grows fast).
Sample Input: 5
Expected Output: 120
Test Case Count: 3

254. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q77: Simple I
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Simple Interest. Read input from stdin and print exact output.
Input Format: Three lines: principal P, rate R, time T (all integers).
Output Format: Print simple interest as integer.
Constraints: Three integers Principal Rate Time whole numbers.
Sample Input: 1000■5■2
Expected Output: 100
Test Case Count: 3

255. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q78: Find Of 3 Numbers
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

256. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q79: Read Until Z
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium

Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

257. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q80: Day Of W itch
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Day Of Week Switch. Read input from stdin and print exact output.
Input Format: One integer.
Output Format: Print abbreviated weekday exactly: Mon Tue Wed Thu Fri Sat Sun
Constraints: Integer 1..7 referencing weekday numbering from handout.
Sample Input: 3
Expected Output: Wed
Test Case Count: 3

258. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q81: Read Until Zero Sum
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

259. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q82: Read Ur Sum
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Read Until Zero Sum. Read input from stdin and print exact output.
Input Format: One integer per line; last line is 0 to stop reading.
Output Format: Print sum of all integers before the line containing 0.
Constraints: User enters positive steps line-by-line stopping at sentinel line reading zero.
Sample Input: 2■12■18■20■0
Expected Output: 52
Test Case Count: 3

260. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q83: Simp est
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Simple Interest. Read input from stdin and print exact output.
Input Format: Three lines: principal P, rate R, time T (all integers).
Output Format: Print simple interest as integer.

Constraints: Three integers Principal Rate Time whole numbers.
Sample Input: 1000■5■2
Expected Output: 100
Test Case Count: 3

261. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q84: Find Largest ers
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Find Largest Of 3 Numbers. Read input from stdin and print exact output.
Input Format: Three integers a, b, c.
Output Format: Print the largest value.
Constraints: Three integers separated on one line (or consecutive reads).
Sample Input: 9 2 5
Expected Output: 9
Test Case Count: 3

262. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q85: Check Ev
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (if, else, switch)'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

263. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q86: Prime Check
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Prime Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Prime or Not
Constraints: One integer roughly ≥2 check primality semantics.
Sample Input: 7
Expected Output: Prime
Test Case Count: 3

264. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q87: Calculat h Menu
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, continue, goto)'.■Problem style: Calculator Switch Menu. Read input from stdin and print exact output.
Input Format: Line 1: op. Line 2: a. Line 3: b. Integer division for '/'.
Output Format: Print integer result.
Constraints: Operator letter then two operands lines.
Sample Input: +■4■11
Expected Output: 15
Test Case Count: 3

265. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q88: Calcu witch Menu
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, L ogical, Assignment'.■Problem style: Calculator Switch Menu. Read input from stdin and print exact output.
Input Format: Line 1: op. Line 2: a. Line 3: b. Integer division for '/'.
Output Format: Print integer result.
Constraints: Operator letter then two operands lines.
Sample Input: +■4■11
Expected Output: 15
Test Case Count: 3

266. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q89: Break On Ne
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement o perators'.■Problem style: Break On Negative Sum. Read input from stdin and print exact output.
Input Format: First line count n, then n integers one per line (last may be negative).
Output Format: Print sum of values before first negative integer.
Constraints: Count prefixed then sequence entries until negative sentinel occurs.
Sample Input: 3■5■8■-2
Expected Output: 13
Test Case Count: 3

267. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q90: Prime Ch
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (i f, else, switch)'.■Problem style: Prime Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Prime or Not
Constraints: One integer roughly ≥2 check primality semantics.
Sample Input: 7
Expected Output: Prime
Test Case Count: 3

268. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q91: Fibonacci Series
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-whi le)'.■Problem style: Fibonacci Series. Read input from stdin and print exact output.
Input Format: One integer n (number of terms).
Output Format: Print first n Fibonacci numbers separated by spaces.
Constraints: Count of terms to emit.
Sample Input: 7
Expected Output: 0 1 1 2 3 5 8
Test Case Count: 3

269. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q92: Print Mu tion Table
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.

Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

270. Phase 2 — Logic Building & Flow Control — Arithmetic, Relational, Logical, Assignment — Q93: Print lication Table
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Arithmetic, Relational, Logical, Assignment'.■Problem style: Print Multiplication Table. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print first 10 multiples of n separated by spaces.
Constraints: One positive integer controlling the factor.
Sample Input: 5
Expected Output: 5 10 15 20 25 30 35 40 45 50
Test Case Count: 3

271. Phase 2 — Logic Building & Flow Control — Increment and Decrement operators — Q94: Check Even/
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Increment and Decrement operators'.■Problem style: Check Even/Odd. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Even if n is even, else Odd.
Constraints: One integer (can be negative or zero).
Sample Input: 8
Expected Output: Even
Test Case Count: 3

272. Phase 2 — Logic Building & Flow Control — Conditional Statements (if, else, switch) — Q95: Leap Yea
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Conditional Statements (if, else, switch)'.■Problem style: Leap Year. Read input from stdin and print exact output.
Input Format: One integer year.
Output Format: Print Leap or Not
Constraints: One calendar year typed.
Sample Input: 2024
Expected Output: Leap
Test Case Count: 3

273. Phase 2 — Logic Building & Flow Control — Loops (for, while, do-while) — Q96: Factorial Of Number
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Loops (for, while, do-while)'.■Problem style: Factorial Of Number. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print n!.
Constraints: Small non-negative integer n (factorial grows fast).
Sample Input: 5
Expected Output: 120
Test Case Count: 3

274. Phase 2 — Logic Building & Flow Control — Jump statements (break, continue, goto) — Q97: Simple I
Module: Phase 2 — Logic Building & Flow Control
Difficulty: medium
Description: Solve a phase 2 — logic building & flow control coding task focused on 'Jump statements (break, c ontinue, goto)'.■Problem style: Simple Interest. Read input from stdin and print exact output.
Input Format: Three lines: principal P, rate R, time T (all integers).
Output Format: Print simple interest as integer.
Constraints: Three integers Principal Rate Time whole numbers.
Sample Input: 1000■5■2
Expected Output: 100
Test Case Count: 3

275. Phase 3 — Functions — Function declaration, definition, and call — Q1: Print Primes In Range
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■ Problem style: Print Primes In Range. Read input from stdin and print exact output.
Input Format: Two integers lo hi on one line.
Output Format: Print all primes in [lo, hi] inclusive, space-separated, ascending.
Constraints: Two bounds inclusive lo hi moderate span.
Sample Input: 5■15
Expected Output: 5 7 11 13
Test Case Count: 3

276. Phase 3 — Functions — Call by value and call by reference — Q2: Factorial Using Function
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Factorial Using Function. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print factorial of n using a function.
Constraints: Same as factorial puzzle but enforce your own helper.
Sample Input: 5
Expected Output: 120
Test Case Count: 3

277. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q3: Armstrong Num k
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial , gcd, lcm'.■Problem style: Armstrong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes or No if n is an Armstrong number.
Constraints: One integer verifying digit-power property.
Sample Input: 153
Expected Output: Yes
Test Case Count: 3

278. Phase 3 — Functions — Call by value and call by reference — Q4: Factorial Using Function
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Factorial Using Function. Read input from stdin and print exact output.

Input Format: One integer n.
Output Format: Print factorial of n using a function.
Constraints: Same as factorial puzzle but enforce your own helper.
Sample Input: 5
Expected Output: 120
Test Case Count: 3

279. Phase 3 — Functions — Function declaration, definition, and call — Q5: Fibonacci Series
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■
Problem style: Fibonacci Series. Read input from stdin and print exact output.
Input Format: One integer n (number of terms).
Output Format: Print first n Fibonacci numbers separated by spaces.
Constraints: Count of terms to emit.
Sample Input: 7
Expected Output: 0 1 1 2 3 5 8
Test Case Count: 3

280. Phase 3 — Functions — Call by value and call by reference — Q6: Sum Using Recursion
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Sum Using Recursion. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print sum of numbers from 1 to n using recursion.
Constraints: One positive integer n.
Sample Input: 5
Expected Output: 15
Test Case Count: 3

281. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q7: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

282. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q8: Armstrong Nun k
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Armstrong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes or No if n is an Armstrong number.
Constraints: One integer verifying digit-power property.
Sample Input: 153
Expected Output: Yes
Test Case Count: 3

283. Phase 3 — Functions — Function declaration, definition, and call — Q9: Factorial Using Function
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■
Problem style: Factorial Using Function. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print factorial of n using a function.
Constraints: Same as factorial puzzle but enforce your own helper.
Sample Input: 5
Expected Output: 120
Test Case Count: 3

284. Phase 3 — Functions — Call by value and call by reference — Q10: Swap Using Pointers
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Swap Using Pointers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped numbers using pointer logic.
Constraints: Two integers as usual swap problem.
Sample Input: 4 9
Expected Output: 9 4
Test Case Count: 3

285. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q11: Lcm Using Gc
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Lcm Using Gcd. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print LCM(a,b) using LCM = a*b/GCD(a,b) with integer arithmetic.
Constraints: Two ints product moderate.
Sample Input: 4■6
Expected Output: 12
Test Case Count: 3

286. Phase 3 — Functions — Call by value and call by reference — Q12: Factorial Using Function
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Factorial Using Function. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print factorial of n using a function.
Constraints: Same as factorial puzzle but enforce your own helper.
Sample Input: 5
Expected Output: 120
Test Case Count: 3

287. Phase 3 — Functions — Function declaration, definition, and call — Q13: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■
Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.

Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

288. Phase 3 — Functions — Call by value and call by reference — Q14: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

289. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q15: Factorial Of Nu
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Factorial Of Number. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print n!.
Constraints: Small non-negative integer n (factorial grows fast).
Sample Input: 5
Expected Output: 120
Test Case Count: 3

290. Phase 3 — Functions — Function declaration, definition, and call — Q16: Fibonacci Series
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■Problem style: Fibonacci Series. Read input from stdin and print exact output.
Input Format: One integer n (number of terms).
Output Format: Print first n Fibonacci numbers separated by spaces.
Constraints: Count of terms to emit.
Sample Input: 7
Expected Output: 0 1 1 2 3 5 8
Test Case Count: 3

291. Phase 3 — Functions — Function declaration, definition, and call — Q17: Lcm Using Gcd
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■Problem style: Lcm Using Gcd. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print LCM(a,b) using LCM = a*b/GCD(a,b) with integer arithmetic.
Constraints: Two ints product moderate.
Sample Input: 4■6
Expected Output: 12
Test Case Count: 3

292. Phase 3 — Functions — Call by value and call by reference — Q18: Armstrong Number Check

Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Armstrong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes or No if n is an Armstrong number.
Constraints: One integer verifying digit-power property.
Sample Input: 153
Expected Output: Yes
Test Case Count: 3

293. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q19: Floyd Triangle rn
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Floyd Triangle Pattern. Read input from stdin and print exact output.
Input Format: One integer n (number of rows).
Output Format: Floyd triangle: row i has i numbers, space-separated; rows separated by newline.
Constraints: Height rows grows consecutive integers pattern.
Sample Input: 3
Expected Output: 1■2 3■4 5 6
Test Case Count: 3

294. Phase 3 — Functions — Call by value and call by reference — Q20: Sum Using Recursion
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Sum Using Recursion. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print sum of numbers from 1 to n using recursion.
Constraints: One positive integer n.
Sample Input: 5
Expected Output: 15
Test Case Count: 3

295. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q21: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

296. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q22: Armstrong Nu ck
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Armstrong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.

Output Format: Print Yes or No if n is an Armstrong number.
Constraints: One integer verifying digit-power property.
Sample Input: 153
Expected Output: Yes
Test Case Count: 3

297. Phase 3 — Functions — Function declaration, definition, and call — Q23: Factorial Using Function
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■ Problem style: Factorial Using Function. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print factorial of n using a function.
Constraints: Same as factorial puzzle but enforce your own helper.
Sample Input: 5
Expected Output: 120
Test Case Count: 3

298. Phase 3 — Functions — Call by value and call by reference — Q24: Swap Using Pointers
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Swap Using Pointers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped numbers using pointer logic.
Constraints: Two integers as usual swap problem.
Sample Input: 4 9
Expected Output: 9 4
Test Case Count: 3

299. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q25: Lcm Using Gc
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Lcm Using Gcd. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print LCM(a,b) using LCM = a*b/GCD(a,b) with integer arithmetic.
Constraints: Two ints product moderate.
Sample Input: 4■6
Expected Output: 12
Test Case Count: 3

300. Phase 3 — Functions — Call by value and call by reference — Q26: Factorial Using Function
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Factorial Using Function. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print factorial of n using a function.
Constraints: Same as factorial puzzle but enforce your own helper.
Sample Input: 5
Expected Output: 120
Test Case Count: 3

301. Phase 3 — Functions — Function declaration, definition, and call — Q27: Gcd Euclidean

Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■ Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

302. Phase 3 — Functions — Call by value and call by reference — Q28: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

303. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q29: Factorial Of N
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Factorial Of Number. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print n!.
Constraints: Small non-negative integer n (factorial grows fast).
Sample Input: 5
Expected Output: 120
Test Case Count: 3

304. Phase 3 — Functions — Function declaration, definition, and call — Q30: Fibonacci Series
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■ Problem style: Fibonacci Series. Read input from stdin and print exact output.
Input Format: One integer n (number of terms).
Output Format: Print first n Fibonacci numbers separated by spaces.
Constraints: Count of terms to emit.
Sample Input: 7
Expected Output: 0 1 1 2 3 5 8
Test Case Count: 3

305. Phase 3 — Functions — Function declaration, definition, and call — Q31: Lcm Using Gcd
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■ Problem style: Lcm Using Gcd. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print LCM(a,b) using LCM = a*b/GCD(a,b) with integer arithmetic.
Constraints: Two ints product moderate.

Sample Input: 4■6
Expected Output: 12
Test Case Count: 3

306. Phase 3 — Functions — Call by value and call by reference — Q32: Armstrong Number Check
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Armstrong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes or No if n is an Armstrong number.
Constraints: One integer verifying digit-power property.
Sample Input: 153
Expected Output: Yes
Test Case Count: 3

307. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q33: Floyd Triangle rn
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Floyd Triangle Pattern. Read input from stdin and print exact output.
Input Format: One integer n (number of rows).
Output Format: Floyd triangle: row i has i numbers, space-separated; rows separated by newline.
Constraints: Height rows grows consecutive integers pattern.
Sample Input: 3
Expected Output: 1■2 3■4 5 6
Test Case Count: 3

308. Phase 3 — Functions — Call by value and call by reference — Q34: Sum Using Recursion
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Sum Using Recursion. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print sum of numbers from 1 to n using recursion.
Constraints: One positive integer n.
Sample Input: 5
Expected Output: 15
Test Case Count: 3

309. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q35: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

310. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q36: Armstrong Nu ck

Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Armstrong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes or No if n is an Armstrong number.
Constraints: One integer verifying digit-power property.
Sample Input: 153
Expected Output: Yes
Test Case Count: 3

311. Phase 3 — Functions — Function declaration, definition, and call — Q37: Factorial Using Function
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■Problem style: Factorial Using Function. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print factorial of n using a function.
Constraints: Same as factorial puzzle but enforce your own helper.
Sample Input: 5
Expected Output: 120
Test Case Count: 3

312. Phase 3 — Functions — Call by value and call by reference — Q38: Swap Using Pointers
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Swap Using Pointers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped numbers using pointer logic.
Constraints: Two integers as usual swap problem.
Sample Input: 4 9
Expected Output: 9 4
Test Case Count: 3

313. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q39: Lcm Using Gc
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Lcm Using Gcd. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print LCM(a,b) using LCM = a*b/GCD(a,b) with integer arithmetic.
Constraints: Two ints product moderate.
Sample Input: 4■6
Expected Output: 12
Test Case Count: 3

314. Phase 3 — Functions — Function declaration, definition, and call — Q40: Fibonacci Series
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■Problem style: Fibonacci Series. Read input from stdin and print exact output.
Input Format: One integer n (number of terms).
Output Format: Print first n Fibonacci numbers separated by spaces.
Constraints: Count of terms to emit.

Sample Input: 7
Expected Output: 0 1 1 2 3 5 8
Test Case Count: 3

315. Phase 3 — Functions — Function declaration, definition, and call — Q41: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

316. Phase 3 — Functions — Call by value and call by reference — Q42: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

317. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q43: Factorial Of Nu
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Factorial Of Number. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print n!.
Constraints: Small non-negative integer n (factorial grows fast).
Sample Input: 5
Expected Output: 120
Test Case Count: 3

318. Phase 3 — Functions — Call by value and call by reference — Q44: Sum Using Recursion
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Sum Using Recursion. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print sum of numbers from 1 to n using recursion.
Constraints: One positive integer n.
Sample Input: 5
Expected Output: 15
Test Case Count: 3

319. Phase 3 — Functions — Function declaration, definition, and call — Q45: Lcm Using Gcd
Module: Phase 3 — Functions
Difficulty: medium

Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■
Problem style: Lcm Using Gcd. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print LCM(a,b) using LCM = a*b/GCD(a,b) with integer arithmetic.
Constraints: Two ints product moderate.
Sample Input: 4■6
Expected Output: 12
Test Case Count: 3

320. Phase 3 — Functions — Call by value and call by reference — Q46: Armstrong Number Check
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Armstrong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes or No if n is an Armstrong number.
Constraints: One integer verifying digit-power property.
Sample Input: 153
Expected Output: Yes
Test Case Count: 3

321. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q47: Floyd Triangle rn
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Floyd Triangle Pattern. Read input from stdin and print exact output.
Input Format: One integer n (number of rows).
Output Format: Floyd triangle: row i has i numbers, space-separated; rows separated by newline.
Constraints: Height rows grows consecutive integers pattern.
Sample Input: 3
Expected Output: 1■2 3■4 5 6
Test Case Count: 3

322. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q48: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

323. Phase 3 — Functions — Function declaration, definition, and call — Q49: Factorial Using Function
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■
Problem style: Factorial Using Function. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print factorial of n using a function.
Constraints: Same as factorial puzzle but enforce your own helper.
Sample Input: 5

Expected Output: 120
Test Case Count: 3

324. Phase 3 — Functions — Call by value and call by reference — Q50: Swap Using Pointers
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Swap Using Pointers. Read input from stdin and print exact output.
Input Format: Two integers a and b.
Output Format: Print swapped numbers using pointer logic.
Constraints: Two integers as usual swap problem.
Sample Input: 4 9
Expected Output: 9 4
Test Case Count: 3

325. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q51: Lcm Using Gc
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Lcm Using Gcd. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print LCM(a,b) using LCM = a*b/GCD(a,b) with integer arithmetic.
Constraints: Two ints product moderate.
Sample Input: 4■6
Expected Output: 12
Test Case Count: 3

326. Phase 3 — Functions — Function declaration, definition, and call — Q52: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

327. Phase 3 — Functions — Call by value and call by reference — Q53: Gcd Euclidean
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Gcd Euclidean. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print GCD(a,b).
Constraints: Two positive ints.
Sample Input: 48■18
Expected Output: 6
Test Case Count: 3

328. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q54: Factorial Of Nu
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial

, gcd, lcm'.■Problem style: Factorial Of Number. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print n!.
Constraints: Small non-negative integer n (factorial grows fast).
Sample Input: 5
Expected Output: 120
Test Case Count: 3

329. Phase 3 — Functions — Function declaration, definition, and call — Q55: Lcm Using Gcd
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Function declaration, definition, and call'.■
Problem style: Lcm Using Gcd. Read input from stdin and print exact output.
Input Format: Two integers a b on separate lines.
Output Format: Print LCM(a,b) using LCM = a*b/GCD(a,b) with integer arithmetic.
Constraints: Two ints product moderate.
Sample Input: 4■6
Expected Output: 12
Test Case Count: 3

330. Phase 3 — Functions — Call by value and call by reference — Q56: Armstrong Number Check
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Call by value and call by reference'.■Problem style: Armstrong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes or No if n is an Armstrong number.
Constraints: One integer verifying digit-power property.
Sample Input: 153
Expected Output: Yes
Test Case Count: 3

331. Phase 3 — Functions — Using functions for primes, powers, factorial, gcd, lcm — Q57: Floyd Triangle rn
Module: Phase 3 — Functions
Difficulty: medium
Description: Solve a phase 3 — functions coding task focused on 'Using functions for primes, powers, factorial, gcd, lcm'.■Problem style: Floyd Triangle Pattern. Read input from stdin and print exact output.
Input Format: One integer n (number of rows).
Output Format: Floyd triangle: row i has i numbers, space-separated; rows separated by newline.
Constraints: Height rows grows consecutive integers pattern.
Sample Input: 3
Expected Output: 1■2 3■4 5 6
Test Case Count: 3

332. Phase 4 — Data Collections — One-dimensional arrays — Q1: Array Reverse Print
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Array Reverse Print. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print elements in reverse order, space-separated.
Constraints: `n` then list length n ints.
Sample Input: 4■1 2 3 4
Expected Output: 4 3 2 1

Test Case Count: 3

333. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q2: Matrix Print 3X3
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Print 3X3. Read input from stdin and print exact output.
Input Format: Three lines, each with three integers.
Output Format: Print matrix row-wise with spaces between numbers per row.
Constraints: Exactly nine ints row major three lines typed.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 1 2 3■4 5 6■7 8 9
Test Case Count: 2

334. Phase 4 — Data Collections — Strings basics and manipulation — Q3: Reverse A String
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Reverse A String. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print reversed string.
Constraints: One word/token without spaces (matching how the checker reads input).
Sample Input: hello
Expected Output: olleh
Test Case Count: 3

335. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q4: Matrix Print 3X3
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Print 3X3. Read input from stdin and print exact output.
Input Format: Three lines, each with three integers.
Output Format: Print matrix row-wise with spaces between numbers per row.
Constraints: Exactly nine ints row major three lines typed.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 1 2 3■4 5 6■7 8 9
Test Case Count: 2

336. Phase 4 — Data Collections — One-dimensional arrays — Q5: Max Min Average Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Max Min Average Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print max min average as integers: average = floor(sum/n).
Constraints: `n` then n ints manageable array sizes.
Sample Input: 4■2 8 4 6
Expected Output: 8 2 5
Test Case Count: 3

337. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q6: Matrix Row Column Sums
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Row Column Sums. Read input from stdin and print exact output.

Input Format: Three lines, each three integers.
Output Format: Line 1: row sums space-separated. Line 2: column sums space-separated.
Constraints: `3 × 3` numeric grid unchanged.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 6 15 24■12 15 18
Test Case Count: 2

338. Phase 4 — Data Collections — Strings basics and manipulation — Q7: Check Palindrome
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Check Palindrome. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print Yes if palindrome, else No.
Constraints: One string with no gaps.
Sample Input: madam
Expected Output: Yes
Test Case Count: 3

339. Phase 4 — Data Collections — Strings basics and manipulation — Q8: Reverse A String
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Reverse A String. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print reversed string.
Constraints: One word/token without spaces (matching how the checker reads input).
Sample Input: hello
Expected Output: olleh
Test Case Count: 3

340. Phase 4 — Data Collections — One-dimensional arrays — Q9: Second Largest Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Second Largest Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print the largest value that is strictly less than the global maximum (ignore duplicate copies of the maximum).
Constraints: At least length two ints general distinct semantics allowed duplicates.
Sample Input: 5■3 9 9 1 4
Expected Output: 4
Test Case Count: 5

341. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q10: Matrix Addition
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Addition. Read input from stdin and print exact output.
Input Format: r c, then matrix A (r lines), then matrix B (r lines).
Output Format: Print A+B matrix row-wise.
Constraints: Dimensions plus two matrices stacked in stdin order.
Sample Input: 2 2■1 2■3 4■5 6■7 8
Expected Output: 6 8■10 12
Test Case Count: 3

342. Phase 4 — Data Collections — Strings basics and manipulation — Q11: Concatenate Without Strcat
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Concatenate Without Strcat. Read input from stdin and print exact output.
Input Format: Line 1 string A; line 2 string B.
Output Format: Print A+B exactly.
Constraints: Two textual tokens sequentially lines.
Sample Input: code■lab
Expected Output: codelab
Test Case Count: 3

343. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q12: Matrix Print 3X3
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Print 3X3. Read input from stdin and print exact output.
Input Format: Three lines, each with three integers.
Output Format: Print matrix row-wise with spaces between numbers per row.
Constraints: Exactly nine ints row major three lines typed.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 1 2 3■4 5 6■7 8 9
Test Case Count: 2

344. Phase 4 — Data Collections — One-dimensional arrays — Q13: Linear Search Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Linear Search Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers. Line 3: target.
Output Format: Print 0-based index of first occurrence, or -1 if not found.
Constraints: Array listing plus target finder.
Sample Input: 5■1 4 7 9 2■7
Expected Output: 2
Test Case Count: 3

345. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q14: Matrix Transpose
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Transpose. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Print transpose (3 lines).
Constraints: `3 × 3` square matrix.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 1 4 7■2 5 8■3 6 9
Test Case Count: 3

346. Phase 4 — Data Collections — Strings basics and manipulation — Q15: String Length Manual
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: String Length Manual. Read input from stdin and print exact output.
Input Format: Single line string (ASCII).

Output Format: Print length without using strlen.
Constraints: Non-empty string without embedded newline simplifying scanner.
Sample Input: coding
Expected Output: 6
Test Case Count: 3

347. Phase 4 — Data Collections — One-dimensional arrays — Q16: Max Min Average Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Max Min Average Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print max min average as integers: average = floor(sum/n).
Constraints: `n` then n ints manageable array sizes.
Sample Input: 4■2 8 4 6
Expected Output: 8 2 5
Test Case Count: 3

348. Phase 4 — Data Collections — One-dimensional arrays — Q17: Count Above Average
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Count Above Average. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Count elements strictly greater than floor(average); average = floor(sum/n).
Constraints: Array ints general distribution.
Sample Input: 5■2 2 2 2 8
Expected Output: 1
Test Case Count: 3

349. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q18: Symmetric Matrix Check
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Symmetric Matrix Check. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Print Yes if symmetric (A[i][j]==A[j][i]), else No.
Constraints: `3 × 3` ints.
Sample Input: 1 2 3■2 4 5■3 5 6
Expected Output: Yes
Test Case Count: 3

350. Phase 4 — Data Collections — Strings basics and manipulation — Q19: Count Vowels Consonants Sp
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Count Vowels Consonants Spaces. Read input from stdin and print exact output.
Input Format: One line: sentence.
Output Format: Print three integers: vowels (aeiou AEIOU), consonants (other letters), spaces.
Constraints: One sentence letters spaces only simplifying classification.
Sample Input: Hi there
Expected Output: 3 4 1
Test Case Count: 3

351. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q20: Matrix Row Column Sums

Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Pr oblem style: Matrix Row Column Sums. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Line 1: row sums space-separated. Line 2: column sums space-separated.
Constraints: `3 × 3` numeric grid unchanged.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 6 15 24■12 15 18
Test Case Count: 2

352. Phase 4 — Data Collections — Strings basics and manipulation — Q21: Check Palindrome
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Prob lem style: Check Palindrome. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print Yes if palindrome, else No.
Constraints: One string with no gaps.
Sample Input: madam
Expected Output: Yes
Test Case Count: 3

353. Phase 4 — Data Collections — Strings basics and manipulation — Q22: Reverse A String
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Prob lem style: Reverse A String. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print reversed string.
Constraints: One word/token without spaces (matching how the checker reads input).
Sample Input: hello
Expected Output: olleh
Test Case Count: 3

354. Phase 4 — Data Collections — One-dimensional arrays — Q23: Second Largest Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style : Second Largest Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print the largest value that is strictly less than the global maximum (ignore duplicate copies of the maximum).
Constraints: At least length two ints general distinct semantics allowed duplicates.
Sample Input: 5■3 9 9 1 4
Expected Output: 4
Test Case Count: 5

355. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q24: Matrix Addition
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Pr oblem style: Matrix Addition. Read input from stdin and print exact output.
Input Format: r c, then matrix A (r lines), then matrix B (r lines).
Output Format: Print A+B matrix row-wise.

Constraints: Dimensions plus two matrices stacked in stdin order.
Sample Input: 2 2■1 2■3 4■5 6■7 8
Expected Output: 6 8■10 12
Test Case Count: 3

356. Phase 4 — Data Collections — Strings basics and manipulation — Q25: Concatenate Without Strcat
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Concatenate Without Strcat. Read input from stdin and print exact output.
Input Format: Line 1 string A; line 2 string B.
Output Format: Print A+B exactly.
Constraints: Two textual tokens sequentially lines.
Sample Input: code■lab
Expected Output: codelab
Test Case Count: 3

357. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q26: Matrix Print 3X3
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Print 3X3. Read input from stdin and print exact output.
Input Format: Three lines, each with three integers.
Output Format: Print matrix row-wise with spaces between numbers per row.
Constraints: Exactly nine ints row major three lines typed.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 1 2 3■4 5 6■7 8 9
Test Case Count: 2

358. Phase 4 — Data Collections — One-dimensional arrays — Q27: Linear Search Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Linear Search Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers. Line 3: target.
Output Format: Print 0-based index of first occurrence, or -1 if not found.
Constraints: Array listing plus target finder.
Sample Input: 5■1 4 7 9 2■7
Expected Output: 2
Test Case Count: 3

359. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q28: Matrix Transpose
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Transpose. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Print transpose (3 lines).
Constraints: `3 × 3` square matrix.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 1 4 7■2 5 8■3 6 9
Test Case Count: 3

360. Phase 4 — Data Collections — Strings basics and manipulation — Q29: String Length Manual
Module: Phase 4 — Data Collections

Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Prob lem style: String Length Manual. Read input from stdin and print exact output.
Input Format: Single line string (ASCII).
Output Format: Print length without using strlen.
Constraints: Non-empty string without embedded newline simplifying scanner.
Sample Input: coding
Expected Output: 6
Test Case Count: 3

361. Phase 4 — Data Collections — One-dimensional arrays — Q30: Max Min Average Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style : Max Min Average Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print max min average as integers: average = floor(sum/n).
Constraints: `n` then n ints manageable array sizes.
Sample Input: 4■2 8 4 6
Expected Output: 8 2 5
Test Case Count: 3

362. Phase 4 — Data Collections — One-dimensional arrays — Q31: Count Above Average
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style : Count Above Average. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Count elements strictly greater than floor(average); average = floor(sum/n).
Constraints: Array ints general distribution.
Sample Input: 5■2 2 2 2 8
Expected Output: 1
Test Case Count: 3

363. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q32: Symmetric Matrix Check
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Pr oblem style: Symmetric Matrix Check. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Print Yes if symmetric (A[i][j]==A[j][i]), else No.
Constraints: `3 × 3` ints.
Sample Input: 1 2 3■2 4 5■3 5 6
Expected Output: Yes
Test Case Count: 3

364. Phase 4 — Data Collections — Strings basics and manipulation — Q33: Count Vowels Consonants Sp
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Prob lem style: Count Vowels Consonants Spaces. Read input from stdin and print exact output.
Input Format: One line: sentence.
Output Format: Print three integers: vowels (aeiou AEIOU), consonants (other letters), spaces.
Constraints: One sentence letters spaces only simplifying classification.
Sample Input: Hi there

Expected Output: 3 4 1
Test Case Count: 3

365. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q34: Matrix Row Column Sums
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Row Column Sums. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Line 1: row sums space-separated. Line 2: column sums space-separated.
Constraints: `3 × 3` numeric grid unchanged.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 6 15 24■12 15 18
Test Case Count: 2

366. Phase 4 — Data Collections — Strings basics and manipulation — Q35: Check Palindrome
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Check Palindrome. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print Yes if palindrome, else No.
Constraints: One string with no gaps.
Sample Input: madam
Expected Output: Yes
Test Case Count: 3

367. Phase 4 — Data Collections — Strings basics and manipulation — Q36: Reverse A String
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Reverse A String. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print reversed string.
Constraints: One word/token without spaces (matching how the checker reads input).
Sample Input: hello
Expected Output: olleh
Test Case Count: 3

368. Phase 4 — Data Collections — One-dimensional arrays — Q37: Second Largest Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Second Largest Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print the largest value that is strictly less than the global maximum (ignore duplicate copies of the maximum).
Constraints: At least length two ints general distinct semantics allowed duplicates.
Sample Input: 5■3 9 9 1 4
Expected Output: 4
Test Case Count: 5

369. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q38: Matrix Addition
Module: Phase 4 — Data Collections
Difficulty: medium

Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Pr oblem style: Matrix Addition. Read input from stdin and print exact output.
Input Format: r c, then matrix A (r lines), then matrix B (r lines).
Output Format: Print A+B matrix row-wise.
Constraints: Dimensions plus two matrices stacked in stdin order.
Sample Input: 2 2■1 2■3 4■5 6■7 8
Expected Output: 6 8■10 12
Test Case Count: 3

370. Phase 4 — Data Collections — Strings basics and manipulation — Q39: Concatenate Without Strcat
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Prob lem style: Concatenate Without Strcat. Read input from stdin and print exact output.
Input Format: Line 1 string A; line 2 string B.
Output Format: Print A+B exactly.
Constraints: Two textual tokens sequentially lines.
Sample Input: code■lab
Expected Output: codelab
Test Case Count: 3

371. Phase 4 — Data Collections — One-dimensional arrays — Q40: Max Min Average Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style : Max Min Average Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print max min average as integers: average = floor(sum/n).
Constraints: `n` then n ints manageable array sizes.
Sample Input: 4■2 8 4 6
Expected Output: 8 2 5
Test Case Count: 3

372. Phase 4 — Data Collections — One-dimensional arrays — Q41: Linear Search Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style : Linear Search Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers. Line 3: target.
Output Format: Print 0-based index of first occurrence, or -1 if not found.
Constraints: Array listing plus target finder.
Sample Input: 5■1 4 7 9 2■7
Expected Output: 2
Test Case Count: 3

373. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q42: Matrix Transpose
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Pr oblem style: Matrix Transpose. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Print transpose (3 lines).
Constraints: `3 × 3` square matrix.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 1 4 7■2 5 8■3 6 9

Test Case Count: 3

374. Phase 4 — Data Collections — Strings basics and manipulation — Q43: String Length Manual
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Prob lem style: String Length Manual. Read input from stdin and print exact output.
Input Format: Single line string (ASCII).
Output Format: Print length without using strlen.
Constraints: Non-empty string without embedded newline simplifying scanner.
Sample Input: coding
Expected Output: 6
Test Case Count: 3

375. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q44: Matrix Row Column Sums
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Pr oblem style: Matrix Row Column Sums. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Line 1: row sums space-separated. Line 2: column sums space-separated.
Constraints: `3 × 3` numeric grid unchanged.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 6 15 24■12 15 18
Test Case Count: 2

376. Phase 4 — Data Collections — One-dimensional arrays — Q45: Count Above Average
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style : Count Above Average. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Count elements strictly greater than floor(average); average = floor(sum/n).
Constraints: Array ints general distribution.
Sample Input: 5■2 2 2 2 8
Expected Output: 1
Test Case Count: 3

377. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q46: Symmetric Matrix Check
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Pr oblem style: Symmetric Matrix Check. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Print Yes if symmetric (A[i][j]==A[j][i]), else No.
Constraints: `3 × 3` ints.
Sample Input: 1 2 3■2 4 5■3 5 6
Expected Output: Yes
Test Case Count: 3

378. Phase 4 — Data Collections — Strings basics and manipulation — Q47: Count Vowels Consonants Sp
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Prob lem style: Count Vowels Consonants Spaces. Read input from stdin and print exact output.

Input Format: One line: sentence.
Output Format: Print three integers: vowels (aeiou AEIOU), consonants (other letters), spaces.
Constraints: One sentence letters spaces only simplifying classification.
Sample Input: Hi there
Expected Output: 3 4 1
Test Case Count: 3

379. Phase 4 — Data Collections — Strings basics and manipulation — Q48: Check Palindrome
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Check Palindrome. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print Yes if palindrome, else No.
Constraints: One string with no gaps.
Sample Input: madam
Expected Output: Yes
Test Case Count: 3

380. Phase 4 — Data Collections — One-dimensional arrays — Q49: Second Largest Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Second Largest Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print the largest value that is strictly less than the global maximum (ignore duplicate copies of the maximum).
Constraints: At least length two ints general distinct semantics allowed duplicates.
Sample Input: 5■3 9 9 1 4
Expected Output: 4
Test Case Count: 5

381. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q50: Matrix Addition
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Addition. Read input from stdin and print exact output.
Input Format: r c, then matrix A (r lines), then matrix B (r lines).
Output Format: Print A+B matrix row-wise.
Constraints: Dimensions plus two matrices stacked in stdin order.
Sample Input: 2 2■1 2■3 4■5 6■7 8
Expected Output: 6 8■10 12
Test Case Count: 3

382. Phase 4 — Data Collections — Strings basics and manipulation — Q51: Concatenate Without Strcat
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Concatenate Without Strcat. Read input from stdin and print exact output.
Input Format: Line 1 string A; line 2 string B.
Output Format: Print A+B exactly.
Constraints: Two textual tokens sequentially lines.
Sample Input: code■lab
Expected Output: codelab
Test Case Count: 3

383. Phase 4 — Data Collections — One-dimensional arrays — Q52: Linear Search Array
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Linear Search Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers. Line 3: target.
Output Format: Print 0-based index of first occurrence, or -1 if not found.
Constraints: Array listing plus target finder.
Sample Input: 5■1 4 7 9 2■7
Expected Output: 2
Test Case Count: 3

384. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q53: Matrix Transpose
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Matrix Transpose. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.
Output Format: Print transpose (3 lines).
Constraints: `3 × 3` square matrix.
Sample Input: 1 2 3■4 5 6■7 8 9
Expected Output: 1 4 7■2 5 8■3 6 9
Test Case Count: 3

385. Phase 4 — Data Collections — Strings basics and manipulation — Q54: String Length Manual
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: String Length Manual. Read input from stdin and print exact output.
Input Format: Single line string (ASCII).
Output Format: Print length without using strlen.
Constraints: Non-empty string without embedded newline simplifying scanner.
Sample Input: coding
Expected Output: 6
Test Case Count: 3

386. Phase 4 — Data Collections — One-dimensional arrays — Q55: Count Above Average
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'One-dimensional arrays'.■Problem style: Count Above Average. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Count elements strictly greater than floor(average); average = floor(sum/n).
Constraints: Array ints general distribution.
Sample Input: 5■2 2 2 2 8
Expected Output: 1
Test Case Count: 3

387. Phase 4 — Data Collections — Two-dimensional arrays (matrices) — Q56: Symmetric Matrix Check
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Two-dimensional arrays (matrices)'.■Problem style: Symmetric Matrix Check. Read input from stdin and print exact output.
Input Format: Three lines, each three integers.

Output Format: Print Yes if symmetric (A[i][j]==A[j][i]), else No.
Constraints: `3 × 3` ints.
Sample Input: 1 2 3■2 4 5■3 5 6
Expected Output: Yes
Test Case Count: 3

388. Phase 4 — Data Collections — Strings basics and manipulation — Q57: Count Vowels Consonants Sp
Module: Phase 4 — Data Collections
Difficulty: medium
Description: Solve a phase 4 — data collections coding task focused on 'Strings basics and manipulation'.■Problem style: Count Vowels Consonants Spaces. Read input from stdin and print exact output.
Input Format: One line: sentence.
Output Format: Print three integers: vowels (aeiou AEIOU), consonants (other letters), spaces.
Constraints: One sentence letters spaces only simplifying classification.
Sample Input: Hi there
Expected Output: 3 4 1
Test Case Count: 3

389. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q1: Pascal Tria Rows
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Pascal Triangle Rows. Read input from stdin and print exact output.
Input Format: One integer n = number of rows.
Output Format: Print Pascal triangle: row i has i integers, space-separated.
Constraints: `n` rows modest combinator growth.
Sample Input: 3
Expected Output: 1■1 1■1 2 1
Test Case Count: 3

390. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q2: Bubble Sort Arra
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Bubble Sort Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Print sorted ascending, space-separated (bubble sort / any stable O(n^2) ok).
Constraints: `n` moderate sortable ints.
Sample Input: 4■4 1 3 2
Expected Output: 1 2 3 4
Test Case Count: 3

391. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q3: Bubble Sort Arra
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Bubble Sort Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Print sorted ascending, space-separated (bubble sort / any stable O(n^2) ok).
Constraints: `n` moderate sortable ints.
Sample Input: 4■4 1 3 2
Expected Output: 1 2 3 4
Test Case Count: 3

392. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q4: Bubble So y

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Bubble Sort Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Print sorted ascending, space-separated (bubble sort / any stable O(n^2) ok).
Constraints: `n` moderate sortable ints.
Sample Input: 4■4 1 3 2
Expected Output: 1 2 3 4
Test Case Count: 3

393. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q5: Check Palindron

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Check Palindrome. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print Yes if palindrome, else No.
Constraints: One string with no gaps.
Sample Input: madam
Expected Output: Yes
Test Case Count: 3

394. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q6: Bubble Sort Arra

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Bubble Sort Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Print sorted ascending, space-separated (bubble sort / any stable O(n^2) ok).
Constraints: `n` moderate sortable ints.
Sample Input: 4■4 1 3 2
Expected Output: 1 2 3 4
Test Case Count: 3

395. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q7: Strong Nu eck

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Strong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes if sum of factorials of digits equals n, else No.
Constraints: Integer moderate digit factorial property.
Sample Input: 145
Expected Output: Yes
Test Case Count: 3

396. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q8: Pattern Printing

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Pattern Printing. Read input from stdin and print exact output.

Input Format: One integer n.
Output Format: Print a right-angle star pattern with n lines.
Constraints: One height n determines how many lines of stars.
Sample Input: 3
Expected Output: *■**■***
Test Case Count: 3

397. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q9: Bubble So y
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Bubble Sort Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Print sorted ascending, space-separated (bubble sort / any stable O(n^2) ok).
Constraints: `n` moderate sortable ints.
Sample Input: 4■4 1 3 2
Expected Output: 1 2 3 4
Test Case Count: 3

398. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q10: Palindror er
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Palindrome Number. Read input from stdin and print exact output.
Input Format: One integer n (no leading zeros except single 0).
Output Format: Print Yes if decimal digits read same forward/backward, else No.
Constraints: Integer decimal digit symmetry checking.
Sample Input: 121
Expected Output: Yes
Test Case Count: 3

399. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q11: Equilateral Star n
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Equilateral Star Pattern. Read input from stdin and print exact output.
Input Format: One integer n (rows of centered star triangle).
Output Format: Each row has leading spaces then stars separated by single space; pattern as sample.
Constraints: Row count pyramid pattern.
Sample Input: 4
Expected Output: *■ * *■ * * *■* * * *
Test Case Count: 3

400. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q12: Check Palindro
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Check Palindrome. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print Yes if palindrome, else No.
Constraints: One string with no gaps.
Sample Input: madam

Expected Output: Yes
Test Case Count: 3

401. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q13: Find Sum ay

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Find Sum Of Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print sum of elements.
Constraints: First n, then exactly n integers on the next input line.
Sample Input: 5■1 2 3 4 5
Expected Output: 15
Test Case Count: 3

402. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q14: Floyd Triangle

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Floyd Triangle Pattern. Read input from stdin and print exact output.
Input Format: One integer n (number of rows).
Output Format: Floyd triangle: row i has i numbers, space-separated; rows separated by newline.
Constraints: Height rows grows consecutive integers pattern.
Sample Input: 3
Expected Output: 1■2 3■4 5 6
Test Case Count: 3

403. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q15: Bubble Sort Ar

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Bubble Sort Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Print sorted ascending, space-separated (bubble sort / any stable O(n^2) ok).
Constraints: `n` moderate sortable ints.
Sample Input: 4■4 1 3 2
Expected Output: 1 2 3 4
Test Case Count: 3

404. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q16: Strong Nu heck

Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Strong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes if sum of factorials of digits equals n, else No.
Constraints: Integer moderate digit factorial property.
Sample Input: 145
Expected Output: Yes
Test Case Count: 3

405. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q17: Pattern Printing

Module: Phase 5 — Problem Practice

Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Pattern Printing. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print a right-angle star pattern with n lines.
Constraints: One height n determines how many lines of stars.
Sample Input: 3
Expected Output: *■**■***
Test Case Count: 3

406. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q18: Bubble S ay
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Bubble Sort Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Print sorted ascending, space-separated (bubble sort / any stable O(n^2) ok).
Constraints: `n` moderate sortable ints.
Sample Input: 4■4 1 3 2
Expected Output: 1 2 3 4
Test Case Count: 3

407. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q19: Palindror er
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Palindrome Number. Read input from stdin and print exact output.
Input Format: One integer n (no leading zeros except single 0).
Output Format: Print Yes if decimal digits read same forward/backward, else No.
Constraints: Integer decimal digit symmetry checking.
Sample Input: 121
Expected Output: Yes
Test Case Count: 3

408. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q20: Equilateral Star n
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Equilateral Star Pattern. Read input from stdin and print exact output.
Input Format: One integer n (rows of centered star triangle).
Output Format: Each row has leading spaces then stars separated by single space; pattern as sample.
Constraints: Row count pyramid pattern.
Sample Input: 4
Expected Output: *■ * *■ * * *■* * * *
Test Case Count: 3

409. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q21: Check Palindro
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Check Palindrome. Read input from stdin and print exact output.
Input Format: One string (no spaces).

Output Format: Print Yes if palindrome, else No.
Constraints: One string with no gaps.
Sample Input: madam
Expected Output: Yes
Test Case Count: 3

410. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q22: Find Sum ay
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and numbe r-theory drills'.■Problem style: Find Sum Of Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print sum of elements.
Constraints: First n, then exactly n integers on the next input line.
Sample Input: 5■1 2 3 4 5
Expected Output: 15
Test Case Count: 3

411. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q23: Floyd Triangle I
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Floyd Triangle Pattern. Read input from stdin and print exact output.
Input Format: One integer n (number of rows).
Output Format: Floyd triangle: row i has i numbers, space-separated; rows separated by newline.
Constraints: Height rows grows consecutive integers pattern.
Sample Input: 3
Expected Output: 1■2 3■4 5 6
Test Case Count: 3

412. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q24: Bubble S ay
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and numbe r-theory drills'.■Problem style: Bubble Sort Array. Read input from stdin and print exact output.
Input Format: Line 1: n. Line 2: n integers.
Output Format: Print sorted ascending, space-separated (bubble sort / any stable O(n^2) ok).
Constraints: `n` moderate sortable ints.
Sample Input: 4■4 1 3 2
Expected Output: 1 2 3 4
Test Case Count: 3

413. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q25: Strong Nu heck
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and numbe r-theory drills'.■Problem style: Strong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes if sum of factorials of digits equals n, else No.
Constraints: Integer moderate digit factorial property.
Sample Input: 145
Expected Output: Yes

Test Case Count: 3

414. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q26: Pattern Printing
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Pattern Printing. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print a right-angle star pattern with n lines.
Constraints: One height n determines how many lines of stars.
Sample Input: 3
Expected Output: *■**■***
Test Case Count: 3

415. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q27: Check Palindro
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Check Palindrome. Read input from stdin and print exact output.
Input Format: One string (no spaces).
Output Format: Print Yes if palindrome, else No.
Constraints: One string with no gaps.
Sample Input: madam
Expected Output: Yes
Test Case Count: 3

416. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q28: Palindror er
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Palindrome Number. Read input from stdin and print exact output.
Input Format: One integer n (no leading zeros except single 0).
Output Format: Print Yes if decimal digits read same forward/backward, else No.
Constraints: Integer decimal digit symmetry checking.
Sample Input: 121
Expected Output: Yes
Test Case Count: 3

417. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q29: Equilateral Star n
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Equilateral Star Pattern. Read input from stdin and print exact output.
Input Format: One integer n (rows of centered star triangle).
Output Format: Each row has leading spaces then stars separated by single space; pattern as sample.
Constraints: Row count pyramid pattern.
Sample Input: 4
Expected Output: *■ * *■ * * *■* * * *
Test Case Count: 3

418. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q30: Strong Nu heck
Module: Phase 5 — Problem Practice

Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and numbe r-theory drills'.■Problem style: Strong Number Check. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print Yes if sum of factorials of digits equals n, else No.
Constraints: Integer moderate digit factorial property.
Sample Input: 145
Expected Output: Yes
Test Case Count: 3

419. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q31: Find Sum ay
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and numbe r-theory drills'.■Problem style: Find Sum Of Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print sum of elements.
Constraints: First n, then exactly n integers on the next input line.
Sample Input: 5■1 2 3 4 5
Expected Output: 15
Test Case Count: 3

420. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q32: Floyd Triangle
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Floyd Triangle Pattern. Read input from stdin and print exact output.
Input Format: One integer n (number of rows).
Output Format: Floyd triangle: row i has i numbers, space-separated; rows separated by newline.
Constraints: Height rows grows consecutive integers pattern.
Sample Input: 3
Expected Output: 1■2 3■4 5 6
Test Case Count: 3

421. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q33: Pattern Printin
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Pattern Printing. Read input from stdin and print exact output.
Input Format: One integer n.
Output Format: Print a right-angle star pattern with n lines.
Constraints: One height n determines how many lines of stars.
Sample Input: 3
Expected Output: *■**■***
Test Case Count: 3

422. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q34: Palindro er
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and numbe r-theory drills'.■Problem style: Palindrome Number. Read input from stdin and print exact output.
Input Format: One integer n (no leading zeros except single 0).
Output Format: Print Yes if decimal digits read same forward/backward, else No.

Constraints: Integer decimal digit symmetry checking.
Sample Input: 121
Expected Output: Yes
Test Case Count: 3

423. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q35: Equilateral Star n
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Equilateral Star Pattern. Read input from stdin and print exact output.
Input Format: One integer n (rows of centered star triangle).
Output Format: Each row has leading spaces then stars separated by single space; pattern as sample.
Constraints: Row count pyramid pattern.
Sample Input: 4
Expected Output: *■ * *■ * * *■* * * *
Test Case Count: 3

424. Phase 5 — Problem Practice — Mixed array, matrix, string, and number-theory drills — Q36: Find Sum ay
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Mixed array, matrix, string, and number-theory drills'.■Problem style: Find Sum Of Array. Read input from stdin and print exact output.
Input Format: First line n, second line n integers.
Output Format: Print sum of elements.
Constraints: First n, then exactly n integers on the next input line.
Sample Input: 5■1 2 3 4 5
Expected Output: 15
Test Case Count: 3

425. Phase 5 — Problem Practice — Patterns, sorts, palindromes, gcd/lcm bundles — Q37: Floyd Triangle
Module: Phase 5 — Problem Practice
Difficulty: medium
Description: Solve a phase 5 — problem practice coding task focused on 'Patterns, sorts, palindromes, gcd/lcm bundles'.■Problem style: Floyd Triangle Pattern. Read input from stdin and print exact output.
Input Format: One integer n (number of rows).
Output Format: Floyd triangle: row i has i numbers, space-separated; rows separated by newline.
Constraints: Height rows grows consecutive integers pattern.
Sample Input: 3
Expected Output: 1■2 3■4 5 6
Test Case Count: 3